Timilehin Adedayo

ML / AI Engineer

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SUMMARY

AI/ML engineer with experience building and deploying production machine learning systems across healthcare, fintech, and agriculture. Contributed to platforms processing thousands of daily inference requests and handling 10,000+ knowledge graph entities. Experienced in ML pipeline architecture, RAG systems, computer vision, and cross-functional team leadership. Founder of CortexAI, where I lead AI development on BRAIN, a medical imaging research project targeting breast cancer diagnosis now in talks for clinical piloting. Open to ML/AI engineering and full-stack AI opportunities.

EDUCATION

Babcock University — Ogun, Nigeria

B.Sc. Computer Science — CGPA: 4.05/5

- Coursework: Machine Learning, Computer Vision, Statistics, Data Structures & Algorithms.

EXPERIENCE

AI Consultant | Acturian — Feb 2026 – Present

Stack: Python, FastAPI, Transformers.

- Built ML pipelines scoring credit risk across 5+ structured and unstructured data sources.
- Boosted fraud classification accuracy ~18% by integrating fine-tuned transformer models into the scoring pipeline.
- Shipped FastAPI microservices handling scoring requests in under 100ms for internal underwriting tools.

Founder & Lead AI Consultant | CortexAI — 2025 – Present

Stack: PyTorch, Swin Transformer, React.

- Founded CortexAI, an AI research company developing BRAIN, a Retrieval-Augmented diagnostic framework for breast cancer detection using mammography imaging.
- Combined a Swin Transformer vision model with a clinical knowledge base to power the diagnosis engine.
- Shrunk model size for edge deployment, enabling the system to run on low-bandwidth devices.

Software Developer Intern | International Institute of Tropical Agriculture (IITA) — Jan – Jun 2025

Stack: Python, TensorFlow, CNNs.

- Trained CNNs to detect cassava disease from field photos, hitting 94% validation accuracy.
- Labeled 12,000 maize crop images with agronomists to train pest-detection models.

ML Engineering Intern | Meridian Analytics — May – Aug 2024

Stack: Python, LSTM, Prophet, Isolation Forest.

- Forecasted building energy use with LSTM and Prophet models across multiple commercial properties.
- Caught equipment failures early using an anomaly-detection system built on Isolation Forest and autoencoders.

SELECTED PROJECTS

BRAIN | Breast Retrieval-Augmented Intelligence Network | CortexAI — Oct 2025

Stack: Python, PyTorch, FAISS, FastAPI, React, Grad-CAM++.

- Designed an explainable AI diagnostic system for breast cancer detection using a Swin Transformer (Swin-B) backbone, achieving 98.99% accuracy on a 4,916-image test set.
- Built a retrieval-augmented verification module using FAISS to cross-check predictions against a 24,000+ image knowledge base.
- Used Grad-CAM++ and Attention Rollout to generate visual heatmaps of pathological features for transparent decision support.

gdg-bu-kg | Enterprise RAG Platform for Academic Data | GDG Babcock — March 2026

Stack: Neo4j Aura, FastAPI, React, Pydantic v2, SQLAlchemy 2.0, PostgreSQL.

- Engineered a RAG platform integrating LLMs with a Neo4j Knowledge Graph (10,000+ academic entities) for high-precision data retrieval.
- Published gdg-bu-kg, a Python SDK with hybrid JWT authentication, reducing query latency by ~500ms.

Yami | P2P Student Lending Platform | Hackathon Finalist, Paystack-Sponsored Ideathon — 2025

- Collaborated in a team build at a Paystack-sponsored ideathon to develop a P2P student lending platform.
- Engineered a Python/FastAPI AI microservice specifically for repayment prediction and adaptive loan structuring.
- Designed a behavioral trust scoring model that served as the core credit infrastructure layer for the application.

Dolphin LLM | Memory-Efficient AI Alignment Research | Independent Study — Oct 2025

Stack: Python, PyTorch, Hugging Face (PEFT, Transformers, Accelerate).

- Fine-tuned a 7B parameter model using LoRA/PEFT on a custom alignment dataset to evaluate safety guardrail degradation.
- Demonstrated that prioritizing honesty over harmlessness led to a 15% reduction in refusal rates for benign boundary-case prompts.
- Documented findings on instruction-following, honesty, and directness trade-offs as part of independent AI safety coursework.

TECHNICAL SKILLS

AI/ML Development:	LLM Fine-tuning (LoRA/PEFT), RAG Architecture, Agentic AI (Autonomous Agents, LangChain, Tool Calling), Computer Vision (CNNs, Vision Transformers, Object Detection), NLP (BERT, NER, Sentiment Analysis), Time-Series Forecasting (LSTM, Prophet), Anomaly & Fraud Detection (GANs, Autoencoders), Explainable AI (Grad-CAM++, SHAP), Knowledge Graphs (Neo4j).
Programming:	Python (FastAPI, SQLAlchemy, Pydantic), JavaScript (React, Node.js), SQL.
Tools:	Neo4j Aura, PostgreSQL, Docker, TensorFlow, PyTorch, Scikit-Learn, GitHub.